Creating Sceneries for the HELI-X V10.1 Model Flight Simulator

English translation of the MCB Bregenz guide (original German text by G. Klauser, 2023)

1. Overview

Ever since model flight simulators first had photo-realistic sceneries, I have been working on the subject. Over the years (with breaks because of work), I have made sceneries for several simulators: Reflex, RealFlight, HELI-X, Aerofly 7/8 and Phoenix. Most were of my club's field, MCB Bregenz, or of my garden or even my basement, so I could practise on the simulator in the most realistic surroundings possible.

During that time I wrote several reports, such as *Sim Sceneries* and *Creating a Scenery for Aerofly 7/8* (see Downloads at klauserg.at).

Over the years the technology and my own skills have developed, and here I want to sum up what I know as of 2023 about the workflow for creating sceneries for model flight simulators.

This report is structured as follows:

- this overview
- capturing panorama photos for flight simulators
- creating the panorama from the individual photos
- creating crash objects
- testing and optimising
- downloads/links

I hope the report is interesting and useful. I would be very happy to get constructive feedback.

2. Capturing the Panorama Photos

Background

There are many very good articles on the web about how to make panoramas. It is well worth searching for and reading a few articles on "creating 360-degree panoramas from individual photos."

Building on that knowledge, I want to describe here what matters most for simulator sceneries, which tools I use today, and what to watch out for when using them.

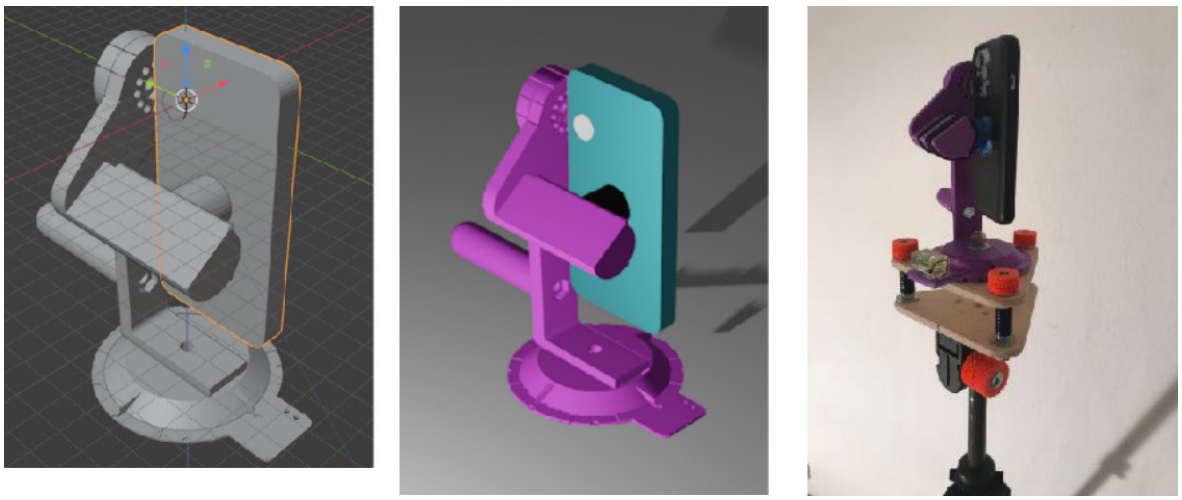
Taking the individual photos for the panorama

- **Choose your usual pilot position as the camera location.** Measure and document the position precisely so the scenery can later be built as realistically as possible. Examples: distance and direction to the centre of the runway, to the safety net, to the tables, or to the windsock. If possible, do not stand too close to the safety net, or it can no longer be shown sharply.
- **Put the camera lens at eye height.** Measure the exact height to the centimetre and write it down.

- **Point the camera so the photos have a clear, known orientation.** For example, the first photo of each "row" faces north, east, south or west (you can find this with the compass on your phone). This makes creating collision objects much easier later.
- **Do not use too wide an angle, but no telephoto either.** About 10-12 photos per 360° with 20-40% overlap works very well. Vertically I use 50% overlap or more, because stitching is made harder by the small number of features in the sky and by fast-moving clouds, and more overlap makes it easier.
- **Keep exposure, focal length and focus constant for all photos.** Take a few test shots of very bright and very dark areas to check for over- or underexposure. Some camera apps also show histograms for this.
- **Get everything in focus.** There are often tables, safety nets etc. near the pilot, so choose exposure settings that keep everything sharp from nearby objects to the distant horizon, without changing the focus.
- **Shoot in RAW if possible.** Otherwise, set a proper white balance.
- **Use a panorama head if possible.** With "crooked" shots, the flying feel is no longer realistic.
- **Check the lens on multi-lens phones.** After finalising all image settings, check which lens is actually taking the picture. The sensor behind it must be at the centre of the panorama head to keep parallax errors to a minimum.
- **Avoid camera shake.** If possible, take photos with a remote trigger or an anti-shake function.
- **Check all your photos before you leave.** Make sure they are complete and good enough for a 360° panorama. A missing or blurred photo can still be retaken now; otherwise you will later have a "hole" in the panorama that makes it unusable.
- **Optionally, take a wide-angle shot of the field.** Most simulators can show a preview image while the scenery loads.

My current equipment

- **Camera:** iPhone 13 Pro
- **Lens used:** 1x (5.7 mm f/1.5)
- **Camera app:** ProCamera
- - very good manual mode, with a histogram for exposure
  - tilt-sensor display for the horizontal and vertical axes in the viewfinder, allowing perfect camera alignment
  - remote trigger via Apple Watch, an anti-shake mode, or a self-timer
  - RAW capture available
- **Panorama head:** home-built from plywood and 3D-printed parts, developed over the years. The phone mounts with a Quad Lock adapter.



The home-built panorama head: CAD model, render and the finished head with phone

3. Creating the Panorama (Stitching and Rendering)

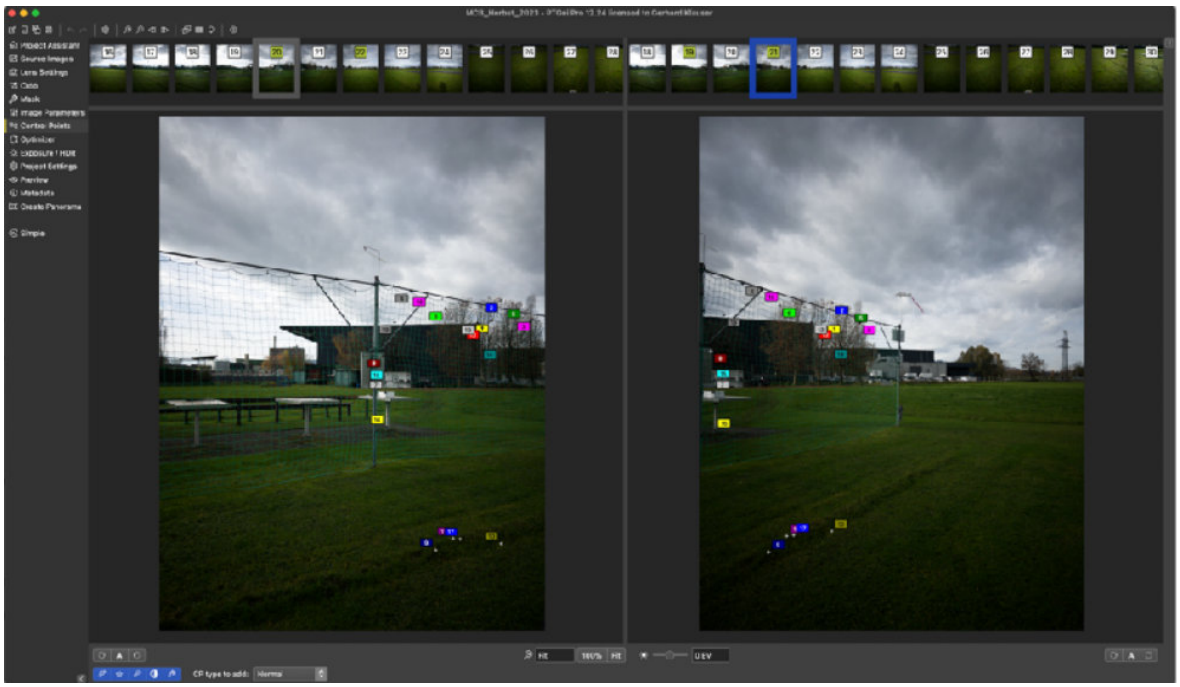
Creating the panorama

An equirectangular panorama (in German, *Plattkarte*) is created from the individual photos. This is a projection of a sphere onto a flat, two-dimensional plane. The following steps are needed:

- **Stitch the photos.** This means telling the program that does the projection how to join the individual photos so that they form a sphere.
- **Render the projection,** correcting the exposure and white balance at the same time.
- The whole process is largely automatic. The better the photos were planned and taken, the less manual work is needed.

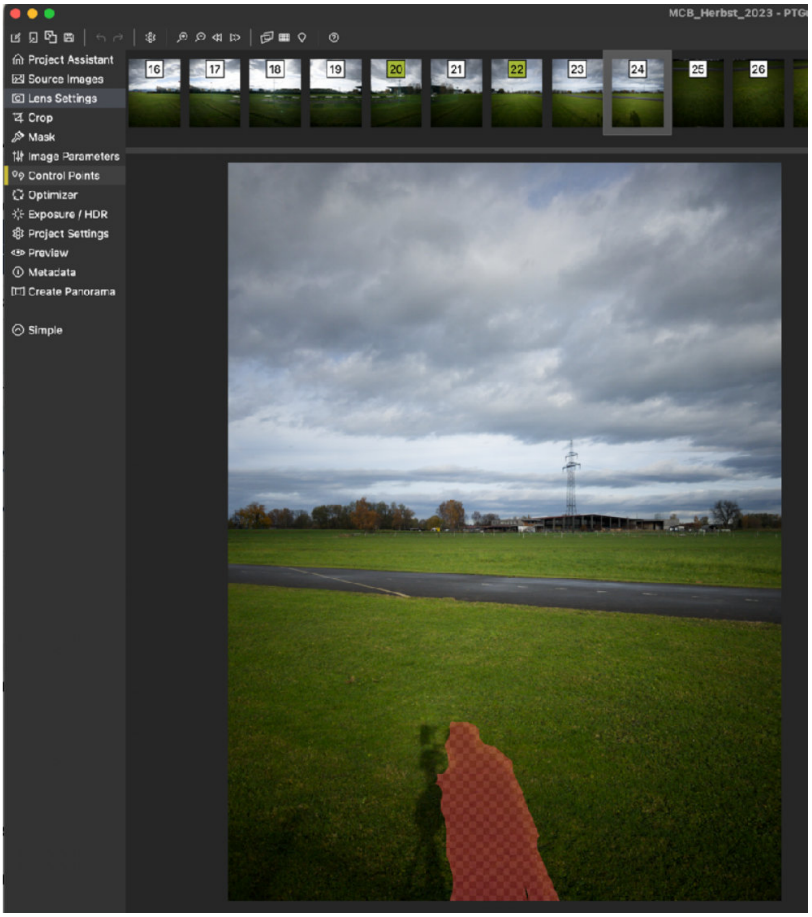
Stitching

- **Control points.** The program creates control points (joining points) for every photo where it can. For each overlapping area, it identifies matching control points in each pair of photos, and the photos are joined at those points.
- - This gets difficult, for example, with a cloudy sky where the clouds move between shots, or with fast objects such as planes or people that appear in different places in a pair of photos. A completely blue or grey sky with no recognisable features is also very difficult.
  - It helps if the horizontal and vertical alignment and the orientation of each photo are known (modern phone cameras store this in the EXIF data).
  - With a completely featureless sky, you can add artificial control points so the photo ends up in the right place in the panorama.



Matching control points between a pair of photos in PTGui

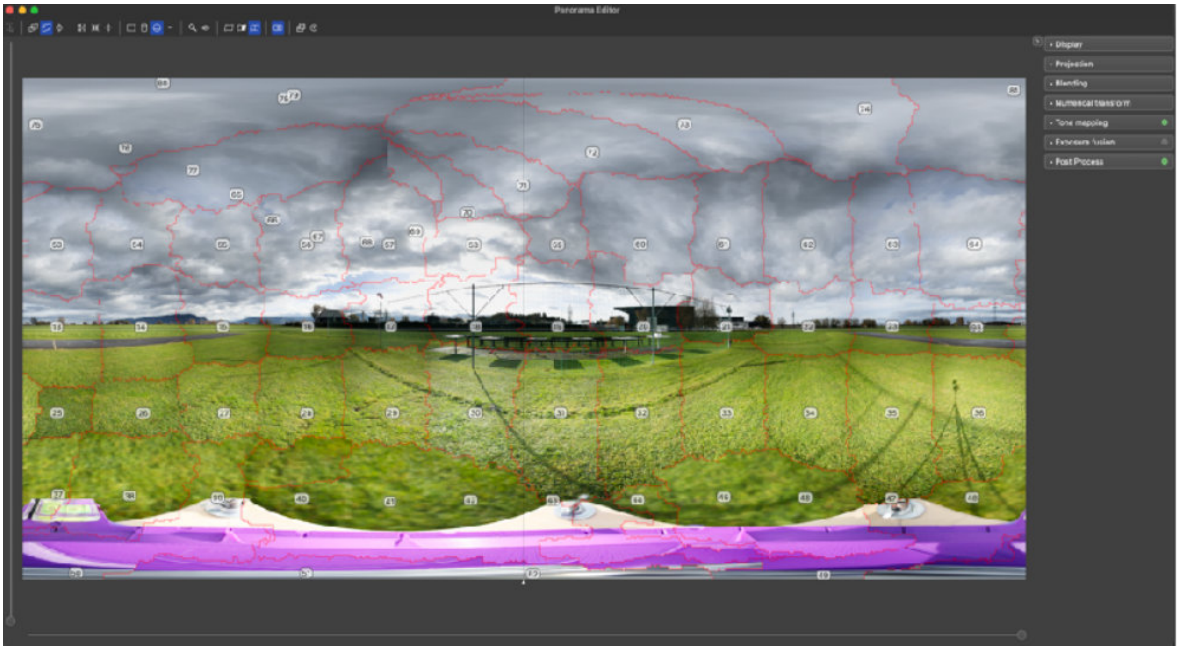
- **Masking.** If individual photos contain objects that do not appear in an overlapping photo, you can mask that area of the photo so the object does not appear in the panorama.



Masking: the photographer's shadow (red) is masked out so it does not appear in the panorama

- **Vertical alignment.** It is also important to align the panorama vertically to the horizon, so you can fly level later and not at a slant. Be careful here if the ground is sloped.

- Our field has a river with a dam. In my first panoramas I used this dam as the horizon. But the dam drops about three metres across the visible area, so the runway, which is actually perfectly level, always looked too short or too long in the simulator, depending on how the panorama was tilted.
  - Today I solve this by finding the horizon with the camera app's tilt sensors. I copy the horizontal row of photos, draw the horizon onto them, and include these extra photos in the stitching. This lets the panorama be aligned horizontally very accurately.
  - When rendering the projection, I then leave out these photos with the drawn horizon so the line does not appear in the panorama.
- **Horizontal orientation.** If you want to add collision objects later, it helps to align the panorama horizontally to a defined compass direction. I use true (geographic) north, because maps you can use for collision objects are normally aligned to true north.
- **Exposure.** When rendering the projection you can still optimise the exposure and apply white balance if needed.
- Finally, render the panorama at the size that suits your simulator.



The panorama editor, showing the seams between the individual photos

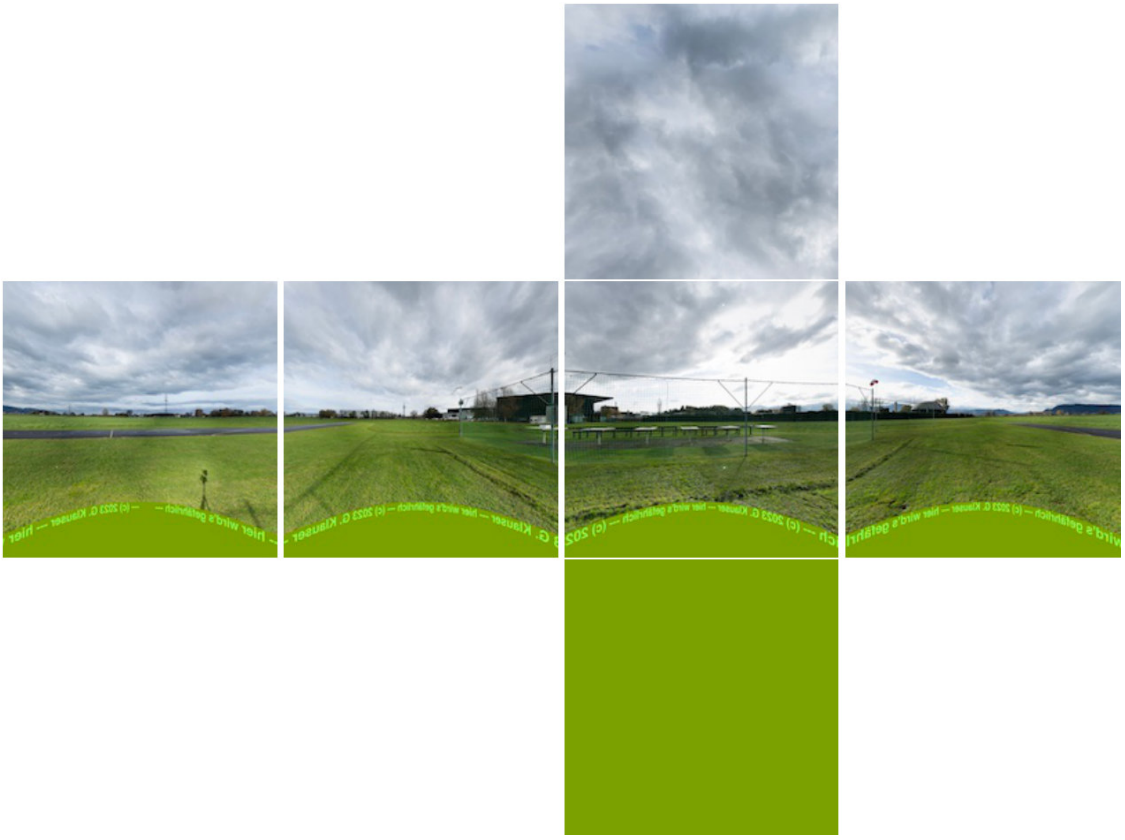
From here on, the next steps depend on the simulator

These are the differences I have come across so far:

- the aspect ratio of the equirectangular projection (Reflex)
- the type of projection: equirectangular or cube (SkyBox)



Equirectangular projection



Cube projection (SkyBox), unfolded

Common stitching programs today all offer the functions needed to convert between these projections.

Extra credit: At the bottom of the panorama you can see the tripod. You do not normally see this while flying, but you can draw something there with a graphics program, such as a copyright notice, the field's coordinates, or a cool slogan.

Next steps

With panoramas made this way, you are basically ready to fly. You just need to add the panorama to the simulator following that simulator's instructions.

- The instructions for Aerofly are here: Aerofly 7/8.
- The instructions for HELI-X are in the Version 10 manual, from section 6.2.1.

But simulator flying only becomes truly realistic when you can also interact with the surroundings (usually by crashing). Creating these crash models is quite a lot of work, but it is definitely worth it. In the next section I will take you through the process of creating crash objects, using HELI-X V10 as the example.

My current equipment

- **Stitcher:** PTGui Pro (paid). Hugin (https://hugin.sourceforge.io/) is a very good free alternative.
- **Image editing:** Pixelmator Pro on my Mac.

4. Creating Crash Objects

What crash objects are and why you need them

In the simulator you are really flying around inside a sphere or a cube (depending on the projection), with the panorama as the background image. So in theory you could fly infinitely far in any direction, even downwards.

Crash objects make the flying experience more realistic and also train your sense of orientation.

The simplest crash object, which every simulator has and which you do not have to set up, is the ground (a plane at 0 m height). If you want to represent uneven terrain, for example for slope soaring, you have to model a new ground. A good example in HELI-X is the "Ahornkopf" airport.

In the description below I will stick to flat sceneries, but the same approach applies to uneven sceneries too.

Crash objects are usually made of surfaces (triangles, polygons). These objects/surfaces have material properties that tell the simulator how to handle them, for example whether they are see-through when you fly behind them (like a wire-mesh fence) or whether they hide the model (like a tree). Details are in the V10 manual, from page 118.

Display in the simulator

Crash objects are normally invisible in the simulator, since they would otherwise hide the panorama image. When creating and testing them, you can show the crash objects with the "Toggle topography display" function. You can assign this function to a key in the control configuration (it is handy to set it in both the helicopter and the airplane control settings).



The black lines show the power line and other objects as crash objects

Another very useful function shows the coordinate grid in the simulator. It is called "Show/hide matrix" and can also be assigned to a key. It lets you work out and estimate directions and dimensions (distances).



The coordinate grid ("matrix") shown in the simulator

If the panorama is very dark, the lines are hard to see. In that case I make a very bright black-and-white version of the original panorama in an image editor and use it for placing the objects. It looks something like this:



Bright black-and-white panorama used while placing objects

When I am finished, I simply switch back to the original panorama.

Level of detail

At the start of my "scenery career" I modelled objects in too much detail: a lot of effort for nothing, and it also slows the simulation down when there are many detailed objects.

Today I model nearby objects with a fairly accurate outline (for example the safety net, where I also model the windsock mounted on it). Distant objects, such as a group of trees, I represent with a single polygon or a "deformed" cylinder rather than individually modelled trees.

Creating the objects

Crash objects are written in XML format. A simple example looks like this:

```
<?xml version="1.0" encoding="utf-8"?>
<project>
  <Triangle FrictionFactor="0.70" CrashSensitivityFactor="1.00" Thickness="1.00" Alpha="0.0000"
    MakeCrashObject="True" ShadowReceiver="False" CeilingType="False" GroundType="False">
    <Point>-10.95,3.30,4.31</Point>
    <Point>-11.21,0.01,4.48</Point>
    <Point>-4.36,3.15,4.15</Point>
    <Normal>0.03,0.02,1.00</Normal>
  </Triangle>
</project>
```

The coordinates in HELI-X are: **X** for left/right, **Y** for up/down and **Z** for front/back. Keep this in mind when placing objects, and especially when using a CAD tool to create them, because there Z is often up/down and Y is front/back.

Fortunately, HELI-X provides a tool that lets you create crash objects in Blender and export them directly to the simulator (see the Blender tool link in section 6). Be sure to read the ReadMe included in the ZIP file; it is very helpful for using the tool.

I measure the size of small objects on site and/or take it from an aerial photo (see Placement). I measure heights where possible and estimate them where measuring is not possible.

I strongly recommend creating a separate Blender object for each crash object, even if several have the same material properties, because this makes testing and optimising much easier.

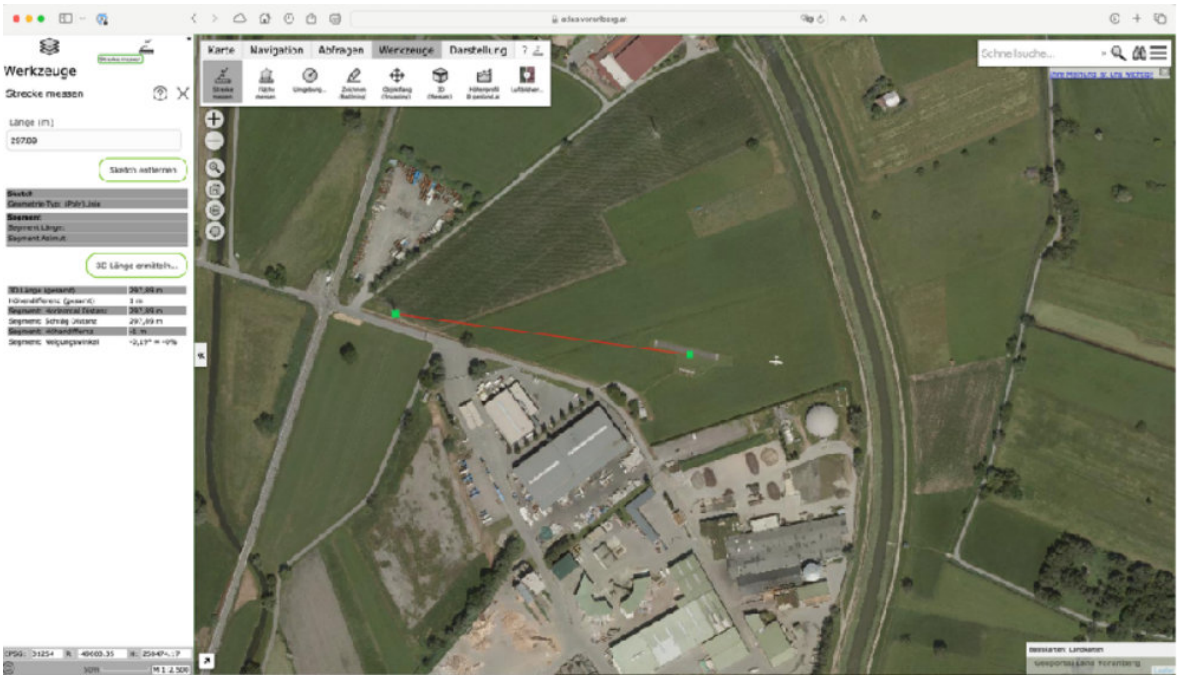
Placement

This is probably the hardest part of the whole job. It is much easier if you followed the instructions for taking the panorama photos:

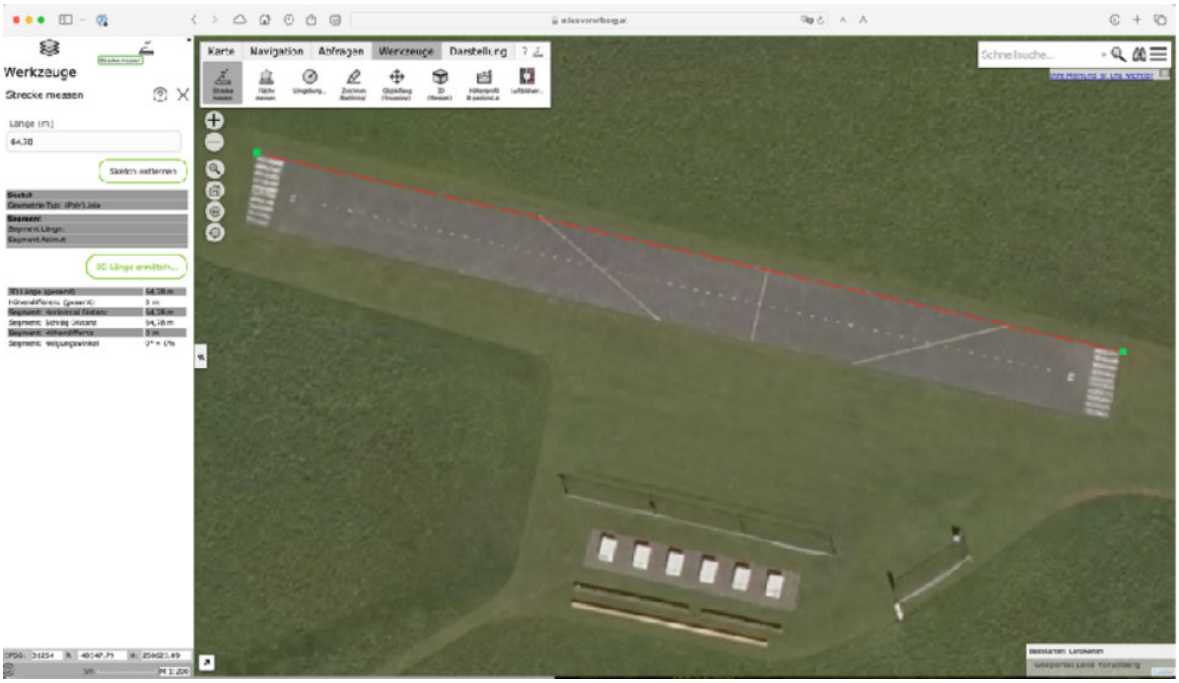
- the height of the lens above the runway level is known
- the camera location is known (GPS coordinates or clear distances to reference objects)
- there are reference objects in the distance for aligning the objects, and possibly nearby ones as a size scale

This approach has worked well for me:

- I download aerial photos from a geo database, for example the Vorarlberg Atlas.
- In the atlas you can measure distances and heights of the terrain, which is extremely useful for checking the scale is correct.

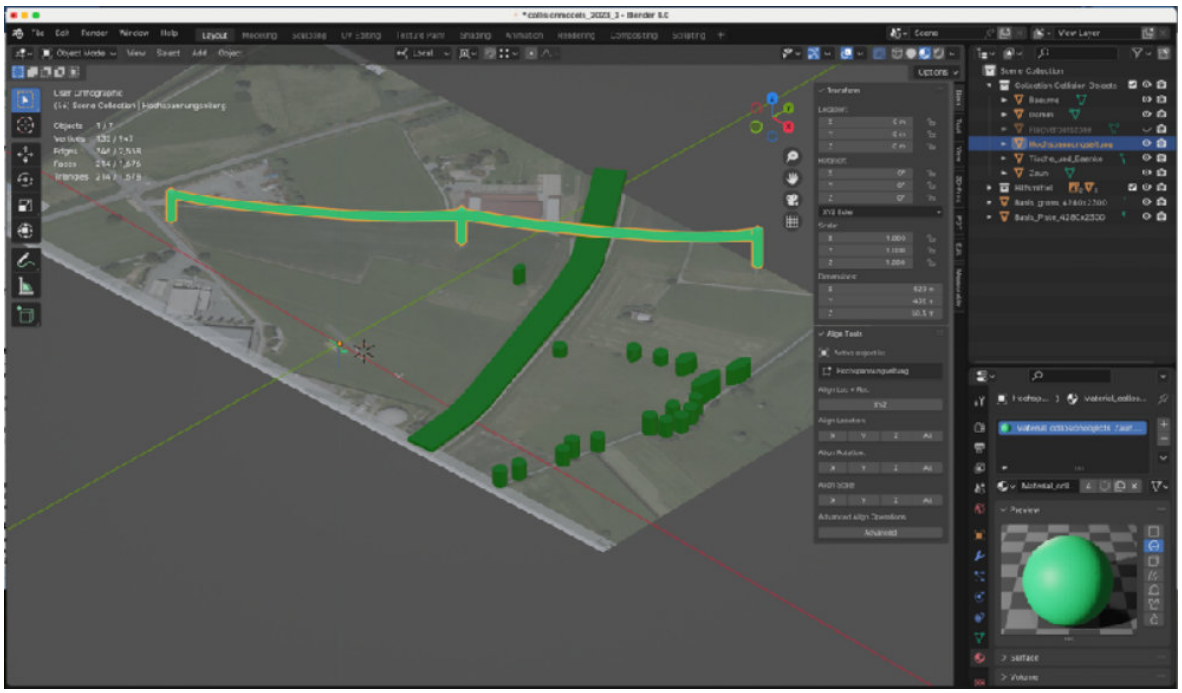


Example: distance from the runway to the WEST power pylon is 297 m, with the ground dropping 1 m

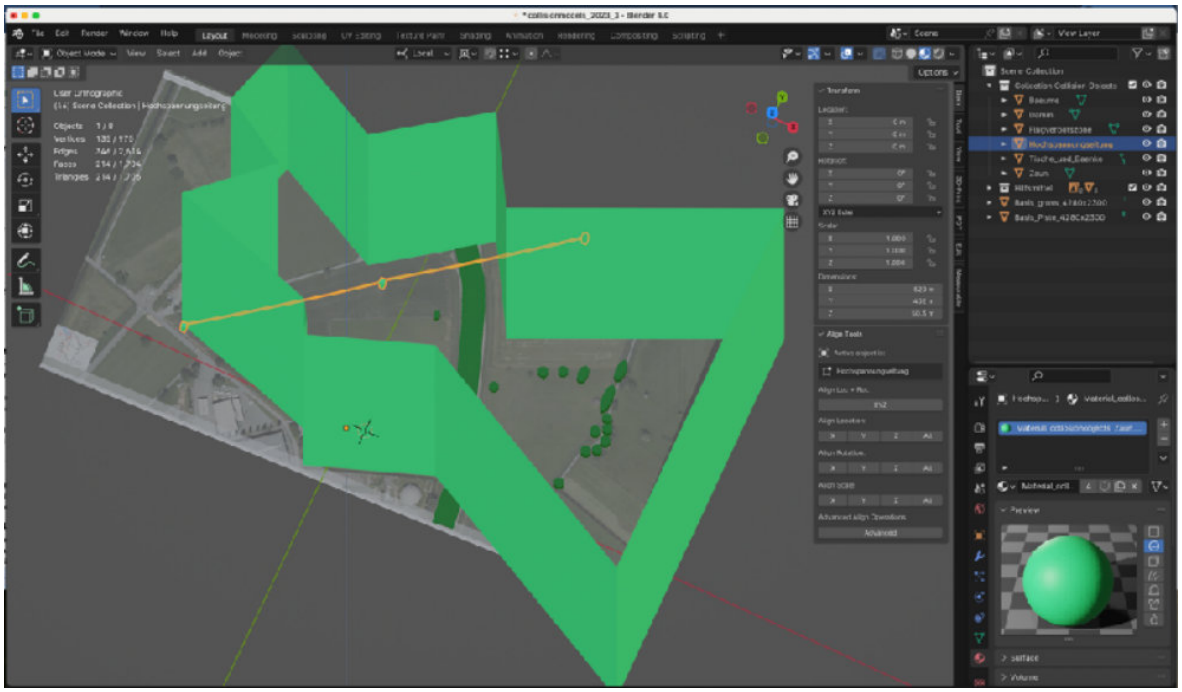


Example: the runway is 64.7 m long with a drop of 0, so it is level

- I load the images into the CAD program (Blender), scale them correctly and place the objects on them. I put the origin of all objects at a common point on the map, e.g. the centre of the runway. This is difficult with vegetation, because the photos are sometimes a bit old and trees or bushes have grown or been cleared since.
- When all objects are created, I rotate the images and objects so they match the orientation of the panorama (the aerial photo, crash objects and panorama must all have the same horizontal orientation). Note that maps are aligned to true north, but a compass (including the one on your phone) shows magnetic north. You can look up the magnetic declination for the location and take it into account when rotating.
- Next, I mark the panorama's camera location in the CAD program and move all objects so that the common origin is at the camera location.



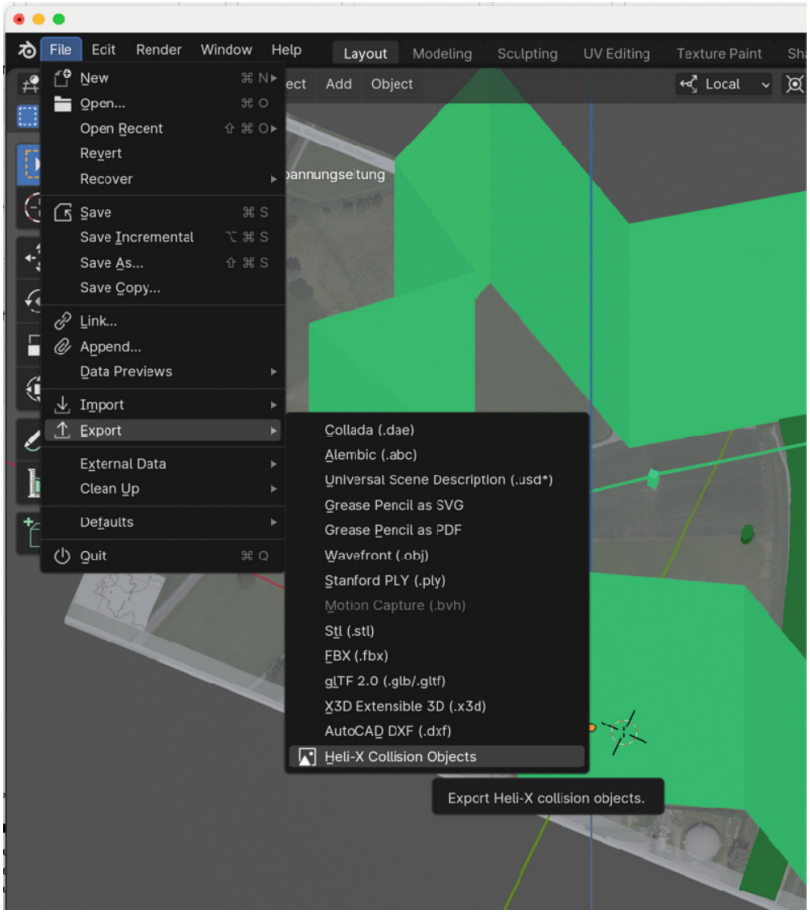
The scene in Blender, without the no-fly zone



The scene in Blender, with the no-fly zone

For MCB members: I moved the no-fly zone behind the fence, so you can fly behind the fence and even land a small model on the table. That is, of course, **strictly forbidden in real life**, but it is enormously helpful for practising orientation in the simulator.

- When everything is finished, export the objects from Blender directly into the HELI-X folder where the scenery is stored (see Integration).



Blender: File > Export > Heli-X Collision Objects

Important: only objects in the Blender collection named "Collection Collision Objects" are exported, and only the ones that are visible.

Integration into the simulator

The airports/sceneries are stored as follows:

- **macOS:** in the user Library under HELI-X10/resources/environments/SkyBox
- **Windows:** in ProgramData\HELI-X10\resources\environments\SkyBox

This is described in the V10 manual from page 116, section 6.2 "Airports".

- Each airport has a folder with the airport's name.
- In the folder are the 6 sides of the cube, which have the same name with the endings _top, _bottom, _left, _right, _back, _front.
- The folder contains an XML file named after the airport (airport name + .xml).
- The crash objects also go in this folder as XML files; you can name them however you like.
- In HELI-X, crash objects are defined with <TopographyFile>.
- In the airport's XML file, add an entry like this for each crash object:

```
.......
<TopographyFile>Baeume.xml</TopographyFile>
<TopographyFile>Damm.xml</TopographyFile>
<TopographyFile>Flugverbotszone.xml</TopographyFile>
<TopographyFile>Hochspannungseitung.xml</TopographyFile>
<TopographyFile>Tische_und_Baenke.xml</TopographyFile>
<TopographyFile>Zaun.xml</TopographyFile>
```

. . . . .

(File names: Trees, Dam, No-fly zone, Power line, Tables and benches, Fence)

Alternatively, you can put all crash objects into one <TopographyFile>. To do this, join the objects in Blender before exporting and export only that one file. I prefer the first option with several files, because it makes developing the objects step by step and debugging easier.

If you want to look at the crash objects of existing airports in Blender, the Blender tool mentioned above can also import them.

Tools

- Any text editor that supports XML, so you spot syntax errors immediately. Better still, Blender with the HELI-X tool.
- Maps or aerial photos, e.g. Vorarlberg Atlas, Google Maps, etc.

5. Testing and Optimising

Testing and optimising

- If you have many objects, it is best to test them one at a time.
- Load the airport with the object you are testing, make the crash objects visible (as described under Creating Crash Objects), and check that the object is in the right place and the shape is right.
- Once that is right, check the behaviour: do you crash when you fly into it, can you see through it, or does it hide the model when you fly behind it, etc.

Symptoms, possible causes and fixes

- **Only one object is out of position, but its size is OK.**
- - Wrong coordinates in the object: correct the coordinates.
- **Several objects are all shifted the same way as seen from the pilot** (e.g. always to the right of where they should be).
- - The panorama's orientation and the objects' orientation do not match: rotate all objects around the pilot position.



In this example, the power line object needs to be rotated to the left

- **Objects on the left and objects on the right are not shifted the same way as seen from the pilot.**

- The pilot position does not match the camera location: move all objects so the pilot position matches them.

- **The distance between distant objects (e.g. pylons or buildings) is OK, but the objects are too big or too small.**

- Too big and too wide: the objects are too close. Increase the scale so the objects move further away.



The object is too big but the spacing between the pylons is OK, so the object may be too close to the pilot

- Too small and too narrow: the objects are too far away. Reduce the scale so the objects move closer to the pilot position.

- Only the height is wrong and the width fits: the height was estimated or measured wrongly. Check and correct it.

- **Nearby objects are too tall and too wide.**

- The distance to the pilot is too small: move the object.



Nearby object too tall and too wide

- **Nearby objects are too low and too narrow.**

- The distance to the pilot is too large: move the object.



Nearby object too low and too narrow

Special case: the camera position was not recorded accurately

If you have a panorama but no accurate data on the camera position, proceed as follows:

- Create the scenery without crash objects.
- In the simulator, turn on the status line, fly to at least 3 reference points and write down their coordinates:



reference_1: x 2.6, y 0.2, z -12.7

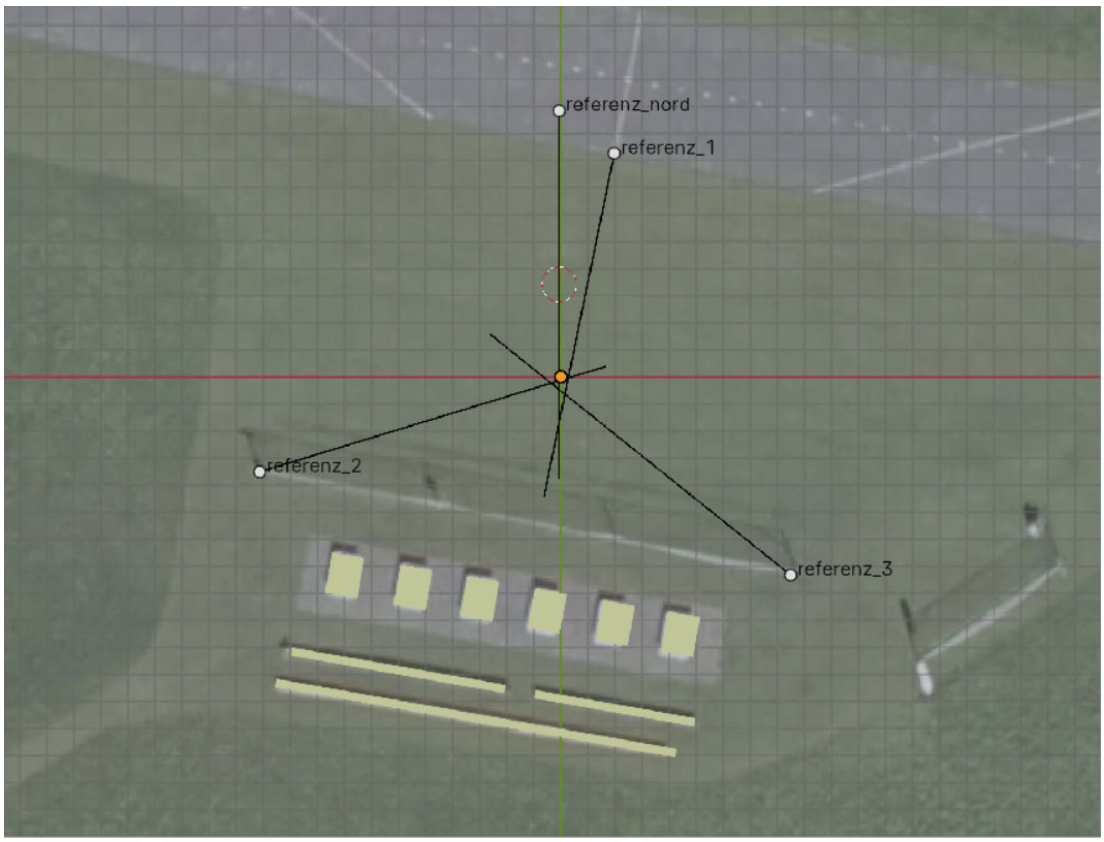


reference_2: x -12.8, y 0.2, z 3.9



reference_3: x 11.1, y 0.2, z 8.9

- In Blender, draw lines from the reference points using the inverted X/Z distances from the reference positions you just recorded.
- The camera position the panorama was taken from is where the lines intersect.
- If the lines are clearly too long, the eye height in the airport XML is too high. If they are too short, the eye height is too low.



The reference lines intersecting at the camera position

- As an extra aid, you can show the matrix, which lets you work out distances and directions (black-and-white lines every 1 metre, red lines every 5 metres).

Final notes on placing objects

Placing objects using aerial photos has a few pitfalls:

- Aerial photos are rarely taken from exactly above the object, so you see a slightly oblique view. This is easy to see in the previous image with the reference points, where you can make out the net.
- The ground is rarely perfectly flat. At our field the ground drops 30-50 cm along the runway, but the runway itself is level. If you draw the runway as an object following the aerial photo, it looks twisted horizontally in the simulator, but it is actually tilted vertically relative to the ground. If you want to correct this, you cannot avoid modelling the terrain accurately.

6. Downloads and Links

- Sceneries and guides: in the download area (klauserg.at)
- HELI-X
- HELI-X manual
- HELI-X Blender tool for collision modelling
- Older report on the subject from 2011
- Camera app: ProCamera
- Stitching programs: PTGui and Hugin
- CAD program: Blender

- Vorarlberg Atlas